Concentrated Solar Desalination Power

A Self-Financing Infrastructure Solution For Global Water Scarcity, Reliable Clean Energy and Mineral Reclamation

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1. Executive Summary

The world is facing three crises: shortage of clean water, high electricity demand and critical mineral availability. The problems are complicated by sea level rise challenging sovereign integrity. Global sea levels are rising at an accelerating rate: currently 4.5 mm/year on average, reaching 5.9 mm/year in 2024, threatening trillions of dollars in coastal infrastructure and displacing hundreds of millions of people. Existing international climate initiatives have produced no concrete, scalable engineering solution to attack this crisis. Freshwater scarcity is reaching catastrophic levels throughout the world. The increase of electric vehicles, Artificial Intelligence data centers, and the need to develop increasingly complex computer chips is outstripping our ability to produce sufficient electricity. Environmentally friendly sources of vital minerals need to be found. Currently, mineral sources are concentrated in a small number of locations, creating dependencies that undermine energy security for the rest of the world. This document describes a compelling investment case to help reduce these critical problems. It proposes a novel integration of subcomponents of existing workable technologies into a unified industrial system that can be profitable through an economy of scale. Concentrated Solar Desalination Power (CSDP) plants, unlike conventional climate proposals that require sacrifice and subsidies, this solution generates revenue from three independent streams: fresh water sales, electricity generation, and mineral extraction. It is environmentally friendly. The energy source, sunlight, is free. There are no fuel costs, no CO_2 emissions, and no hazardous waste. A pilot program is proposed to demonstrate CSDP viability and support the development of a critically needed backup TSMC chip manufacturing plant in Arizona.

2. The Problem: Converging Crises

2.1 Fresh Water Scarcity: A Human Rights Imperative

The United Nations recognized access to safe drinking water and sanitation as a human right in 2010 (Resolution 64/292). Yet as of 2026, more than 2 billion people lack reliable access to safe water, and projections indicate that global demand will exceed available supply by 40 percent by 2050. The burden falls disproportionately on the Global South: Sub-Saharan Africa, South Asia, the Middle East, and North Africa - precisely the regions with the greatest solar energy potential and the closest proximity to saltwater coastlines.

Fresh water scarcity is already a driver of conflict, migration, and state fragility. The UN Secretary-General has repeatedly identified water stress as a multiplier of geopolitical instability. Addressing it at scale is not merely a development objective, it is a peace and security imperative.

2.2 Growing Clean Energy Requirements: Falling behind

The global transition to clean energy is aggravated by the increasing electrification of transport and the rapid expansion of digital infrastructure. A huge expansion of a resource in critically short supply is mandatory: reliable baseload electricity.

2.3 Critical Minerals Availability: The Equity Dimension

The minerals, lithium, magnesium, potassium, and rare earth elements are used in battery storage, agricultural inputs, AI data centers, and advanced computer chip fabrication. Current supply chains for these minerals are concentrated in a small number of states, creating dependencies that undermine energy security for the majority of Member States. Seawater contains commercially significant concentrations of all of these minerals. A global desalination network would, as a byproduct of its primary mission, distribute mineral extraction capacity across all participating coastal nations - democratizing access to resources currently monopolized by geography.

3. The Solution: The Concentrated Solar Desalination Power Plant

3.1 How It Works

The following is a first of its kind proposal that describes a unique solution to the crises noted above: The Concentrated Solar Desalination Power (CSDP) plant. The design incorporates several special manufacturing techniques and inexpensive materials to speed construction, improve profitability, and achieve economy of scale.

1. Seawater is pumped from coastal tidal sources through plastic pipelines manufactured on-site. Using 3-D printing on site will substantially reduce material and construction costs and potential supply chain dependancies or disruptions. A huge amount of plastic would be needed for this project on a global scale. Recycled plastic could be used where possible, helping reduce a significant waste disposal problem for the environment. Plastic obtained from petrochemical companies would encourage their support for the plan to offset their loss of revenue from gasoline production. From an environmental standpoint, the carbon foot print is reduced when carbon is sequestered in plastic.

2. During sunlight periods concentrated solar mirrors focus sunlight on heliostats which in turn heat salt to 700°C. A heliostat is a device with one or more flat mirrors that continuously track the sun and reflect sunlight toward a specific point, compensating for the sun's motion across the sky. The term comes from Greek: helios (sun) and stat (stationary). The mirror doesn't keep the sun stationary - it keeps the reflected beam stationary on the target. Thousands of heliostats form a heliostat field that focuses sunlight onto a central receiver atop a tower melting the salt inside.

3. The molten salt flows to underground storage vats to conserve the heat. This provides continuous heat to reliably create steam from the seawater 24 hours a day, regardless of cloud cover or nighttime. This provides for true 24-hour power generation and desalination, unlike conventional solar photo voltaic or wind driven systems. It also eliminates the green house gas emissions emanating from carbon-fired plants.

4. Molten salt heats the incoming seawater, piped underground, to create steam. This drives turbine generators producing electricity. These turbines are standard, off-the-shelf variety.

5. The desalinization process requires no external source of energy. Steam exiting the turbines is first pre-cooled with incoming seawater and then further condensed by the environment as it is distributed from the plant to point of use producing distilled fresh water. The incoming seawater is pre-heated by the steam to recover waste heat. This efficiently saves heat needed from the molten salt to create the steam.

6. The fresh water is pumped inland via plastic pipelines to aquifers, reservoirs, farms, and industrial and population centers using electricity produced by the plant. Once up and running, plant electricity is used to bring seawater to the plant.

7. Initial brine discharge will be reinjected into a secondary desalinization process for further concentration. When highly concentrated, the residual brine is processed for commercial mineral extraction: sodium chloride, magnesium, potassium, calcium, sulfate, and trace lithium, creating an additional revenue stream. As seawater concentrates during evaporation, minerals naturally crystallize in predictable stages. Calcium carbonate and gypsum precipitate first, followed by sodium chloride, magnesium salts, and potassium salts. This staged crystallization allows controlled harvesting of industrial minerals while progressively reducing the brine volume.

Large desalination plants face political opposition and permitting obstacles because of brine discharge into oceans. Thus the equally important result of CSDP mineral extraction is that it minimizes the brine disposal problem, a principle environmental liability that plagues conventional desalination technologies.

8. Electricity generated from the steam turbines powers the plant and pumps sea water inland. The excess power is sold to regional grids. The number and size of standard generators would depend on the electrical output and desalinization volume desired.

9. The overall plant design and its location, construction, operation, and control systems will be optimized through the use of Artificial Intelligence (AI).

3.2 Similar Systems

Subsystems of the proposed CSDP plant have already been been demonstrated reducing technological risk:

1.Concentrated solar heat has been used to drive steam turbines to produce electricity. Molten-salt tower plants have been used demonstrate high-temperature steam generation. Modern molten-salt tower plants such as Crescent Dunes, Noor Energy 1, and Spain's Gemasolar use heliostats to heat molten salt to 500 to 600°C. for 24-hour operation This stored heat is then transferred to water in a steam generator, producing high-pressure steam for electricity production.

However, they do not validate the full CSDP concept because they are single-output electricity plants. The proposed CSDP architecture argues for integrating desalination of seawater, steam-powered turbine generators, mineral extraction, and inland water transport into a profitable, unified system.

2. Concentrated solar heat has been used evaporate steam for desalination. Sunvapor, a U.S. DOE-funded program, produced a solar steam collector system that concentrates sunlight to generate high-temperature steam for industrial processes including desalination. Their system is explicitly designed to replace

fossil-fuel steam with solar steam. This is a validated solar-thermal steam generation system, though not yet a full-scale CSDP desalination plant.

The Solar-based Steam powered with Salt (SbSpS) Desalination Plant (SME) uses an array of reflectors to concentrate solar radiation onto a molten-salt thermal mass. Seawater is pumped through ceramic tubes embedded in the thermal mass, generating saturated water vapor (steam). The steam drives a turbine and then condenses into fresh water. It is not yet a commercial plant, but it is a complete steam-evaporation desalination design.

3. Concentrated solar heat has been used to extract minerals from seawater, but only in partial, experimental, or hybrid systems. The existing evidence shows that solar-thermal evaporation can concentrate brines enough for mineral recovery, and that solar-powered extraction of specific minerals is technically validated, but the field is still emerging. A major review in 2016 in *Environmental Science: Water Research & Technology* concluded that seawater contains economically significant quantities of valuable minerals and that extraction is technically feasible, though not yet widely commercialized. The challenge is energy cost, which CSDP solves by providing free heat.

4. The largest desalinization plant in the world is the Ras Al-Khair Power and Desalination Plant in Saudi Arabia. It is a hybrid desalination facility, with a desalination capacity on the order of 1.03 million m^3 per day using reverse osmosis. The Ras Al-Khair complex is an integrated water-and-power (IWPP) facility: it includes a large combined-cycle power plant generating roughly 2,400 MW that supplies electricity and low-grade thermal energy to the desalination units. So energy comes from on-site power generation rather than an external grid alone.

The power plant is gas-fired using combined-cycle gas turbines, so it has a large carbon footprint. For desalinization, the facility uses three processes: two thermal distillation processes, Multi-Stage Flash (MSF)and Multi-Effect Distillation (MED), and traditional Reverse Osmosis (RO).

MSF heats seawater and flashes it to steam in successive lower-pressure stages. Its drawbacks are that it has a very high thermal energy demand and capital cost, and works best only when waste heat or cheap thermal energy is available.

The MED process reuses vapor heat across multiple effects to improve thermal efficiency. Its drawbacks are that it is still heat-intensive, susceptible to scaling and corrosion, and requires skilled operation and maintenance.

The other desalinization process, RO, uses high pressure and thin-film composite membranes. It is widely deployed and lowest electrical energy per cubic meter for seawater among mature technologies. High-pressure pumps force seawater through semipermeable membranes that block impurities. Its drawbacks are that membranes tear, foul and require frequent cleaning and continual replacement. It produces concentrated brine that must be managed and its energy costs are significant and fluctuate with the market. The seawater must be filtered and pretreated to remove particles, organics, and microbes that would foul the membrane or cause coagulation. Modern RO plants use devices such as isobaric pressure exchangers to capture the pressure energy in the brine stream and transfer it to incoming feed water. This can reduce energy consumption by up to about 40 percent.

The concentrated brine is approximately roughly twice the original salinity. Brine, along with the cooling condensate streams produced is discharged to the adjacent seawater via engineered outfalls. This coastal desalination plant uses submerged outfalls and mixing zones to dilute and try to disperse the brine before it reaches sensitive nearshore areas, but the discharge still significantly threatens marine environments.

All the technologies described above are proven, some commercially operational, and much of the equipment is available off the shelf. However, they have never been combined into the unified CSDP

process proposed in this paper. No new technologies or exotic inventions are required - only scale and innovative engineering.

4. The Path to Global Adoption: The Mass Production Imperative

The current construction cost of a large-scale desalination facility is approximately $5 to 8 billion. However, as construction of CSDP plants increase, industrial scaling will begin to lower plant costs. Effects of mass production is not an optimistic assumption - it is the documented behavior of every energy technology humanity has scaled to this level. Solar panel costs fell 90 percent in a decade due to standardization and mass production. Wind turbine and battery costs followed identical curves.

The world wide shortage of clean water, electricity, and minerals should encourage increasing global adoption of CSDP as the benefits become more obvious. Simultaneous installations are thus possible, and the demand could rapidly ramp up mass production of components and quickly create economy of scale reductions in costs faster than expected.

As CSDP would likely be an attractive investment opportunity for private capital, costs to governments will be reduced. In addition, as the plants are installed throughout the world, development costs would be shared globally, not borne by only a few countries. Least Developed Countries could donate land to help pay for their plants, further reducing costs.

4.1 Standardization Benefits

The design for plants is proposed to be along the lines of a “cookie cutter” approach. A large number of plants built to a single standardized modular design would benefit from component manufacturing at automotive industry scale - potentially reducing per-plant costs by 60 to 80 percent. The proposal is designed for a learning curve similar to solar, PV, wind, and batteries, where standardized designs and manufacturing scale drive large cost declines over time.

The characteristics of CSDP lends itself to extensive standardization of components such as turbines, control systems, heliostats, mirrors, piping, molten salt storage, fundamental desalinization and demineralization processes, and location requirements.

CSDP components are easy to produce, widely available, require no exotic materials. Plants pose no unusual construction demands or environmental constraints. 3-D printing of plastic pipe infrastructure and plant components on-site from domestic and ocean-sourced recycled plastic further reduces material costs. Continuing advances in AI will streamline plant design, component development, and construction techniques.

4.2 Heliostats

Modern heliostat costs are typically expressed in dollars per square meter of reflective area. The most authoritative source is the U.S. National Renewable Energy Laboratory (NREL), which tracks commercial and advanced heliostat costs. Current commercial cost range $120–$180 per m² for today’s commercial heliostats.This includes:

- Mirror panels
- Steel support structure
- Tracking motors
- Control systems
- Installation and calibration

The U.S. Department of Energy’s Heliostat Consortium (HelioCon) is working to reduce costs to:
$50 per m² This target is aggressive but technically plausible with mass production, lighter materials, and automated manufacturing. A large CSDP plant t may require 10,000 to 20,000 heliostats, totaling 1–2

million square meters of mirror area. At $50/m², the entire solar field would cost $50 to $100 million, which is a small fraction of the total CSDP plant cost.

4.3 Mirrors

Making heliostat mirrors from solar-melted sand is possible in principle but not yet in practice. Sand (silica) melts at approximately 1700°C, and concentrated solar furnaces can exceed 3000°C. Solar furnaces like the one at Odeillo in France already melt metals and ceramics using only sunlight.

The physics for solar melting is straightforward: CSDP heliostats would concentrate sunlight to a solar furnace which melts silica. Molten silica is cast into glass sheets, and glass sheets are then polished and coated into heliostat mirrors.

This approach has not been done commercially (yet) because industrial glassmaking is already extremely optimized and large float-glass factories produce mirror-grade glass at very low cost using fossil heat. Solar melting would be intermittent unless paired with molten salt storage. Molten salt or ceramic storage would be necessary to run a glass plant continuously. Mirror quality requires precise thermal control and optical-grade flatness and low-iron purity are difficult to achieve with variable solar input.

Economics haven't justified the switch to solar furnaces so they usually add to green house gases. Glass is cheap and the expensive part of a heliostat is the structure and tracking system. But the idea is strategically important: for a global-scale CSDP plants, solar-melted glass becomes attractive because it eliminates fossil fuels from glass production and would allow on-site mirror manufacturing in desert regions. It reduces supply-chain risk and would use abundant local sand. On site glass production aligns with CSDP's "closed-loop solar industrialization" vision. This would be especially relevant for North Africa, the Middle East, Australia, the U.S. Southwest, and Atacama, Chile. All have abundant silica sand and high solar flux.

4.4 Massive Cost Savings : A CSDP Manufacturing complex

With global expansion of demands, an enormous amount manufacturing capability, material, and products will be required. A powerful long-term implication of this architecture is that early CSDP plants could mass produce heliostats and their mirrors onsite for future CSDP plants. An early CSDP complex could include: a solar furnace, a molten-salt thermal storage system, a glass-melting and casting line, a mirror-coating facility, and a heliostat assembly yard. This would allow local, solar-powered manufacturing of heliostats, reducing costs and enabling exponential scaling. This is how CSDP becomes a self-replicating industrial system - a concept with enormous strategic implications.

5. The Revenue Model: Three Independent Profit Streams

A defining characteristic of this proposal, and its most important distinction from conventional climate infrastructure, is that it generates profit. Each plant operates as a vertically integrated utility producing three salable commodities with zero fuel cost. The attractive profitability of the model means this infrastructure can be financed as private capital investment project bonds, infrastructure equity, or sovereign wealth funds rather than strictly government funding.

Revenue Streams Market Drivers:

1. Competitive Advantage: Fresh Water: Over 2,000,000,000 people lack safe water. The demand keeps growing - 40 percent by 2050.

2. No fuel cost; only marginal operating expense.

3. Electricity produced 24 hours a day. Electric Vehicle (EV) charging and computing require baseload clean power. Molten salt storage enables round-the-clock generation.

4. Minerals lithium, magnesium, potassium, are available in seawater in sufficient quantities to make their extraction profitable. But the profit is not as valuable as providing those critical resources for for batteries, computer chip production and agriculture. Seawater is an infinite, free mineral feedstock requiring no international aid or government expenditure.

5. Brine disposal has been a major environmental and regulatory obstacle for conventional desalination. Significantly, there is only a minimal brine disposal problem or cost because of the mineral extraction process. All brine is recycled as mineral feedstock, reducing waste streams and converting a liability into viable revenue source.

5.1 Profitability: Contracted Cash Flow and Upside

Output	Contract Type	Cash Flow Profile	Profitability Driver
Water	Municipal/industrial purchase agreements	Stable, inflation-indexable	Scarcity pricing + long-duration demand
Electricity	24/7 baseload PPAs (utilities, AI, Semiconductor fabrication facilities)	Infrastructure-grade yield	High capacity factor and reliability premium
Minerals	Production agreements and spot sales	Diversifying upside stream	Commodity upside and strategic supply premium

CSDP's investment case is strengthened by profit stream diversification: even if one market temporarily weakens (e.g., electricity prices), the other two (water scarcity and strategic minerals) can help sustain debt service and equity returns. This supports project-finance structures with long-term contracted revenues and scalable replication once a first pilot program proves solid economics.

This proposal is not a government aid program.
It is a globally scalable, vertically integrated infrastructure business
with a multi-trillion dollar addressable market and a planetary environmental upside.

6. A Strategic Pilot Program: The Arizona Semiconductor Corridor

6.1 The AI and Semiconductor Infrastructure Imperative

Beyond climate adaptation, CSDP can directly support the physical foundations of the AI economy. AI data centers and leading-edge semiconductor fabrication facilities require continuous, high-capacity electricity, rare minerals, and cooling water. CSDP uniquely produces all three: uninterrupted power, very large volumes of ultrapure, distilled water, and resilient material supply chains for rare minerals making it a potential solution for enabling both national and global security and industrial durability.

6.2 Strategic Value of a Second TSMC Plant

Taiwan Semiconductor Manufacturing Company (TSMC) is widely regarded as the supreme high-volume computer chip producer in the world and is currently the only manufacturer with the technology to make the most advanced computer chips. Nvidia is by far the dominant developer of chips used for global AI development and processing. TSMC makes virtually all Nvidia chips. While Taiwan is not the only place where computer chips are made, including Nvidia's, loss of TSMC production capacity due to sea level rise, drought, power shortages, pandemics, or geopolitical instability would represent a catastrophic

global economic shock to world economies and the modern way of life. A Chinese blockade or invasion would cause an immediate global supply shock, potentially triggering the worst economic downturn since the Great Depression. While not imminent, it was reported in the New York Times on February 24, 2026 by Tripp Mickle that U.S. intelligence officials have warned tech executives, including Nvidia CEO Jensen Huang, that China could be prepared to move against Taiwan as early as 2027.

6.3 Basis for Including Arizona in the Pilot Program

It is imperative that a redundant source of semiconductor fabrication facilities be built as soon as possible to reduce international tensions. Currently, TSMC is planning a backup plant for the Phoenix, Arizona Semiconductor Corridor, but the region faces severe water shortages, electrical grid strain, and mineral insecurity. This makes Arizona an ideal location not only for an immediate proof of concept demonstration for the CSDP concept but also at the same time providing critical support for TSMC's much needed backup plant .

6.4 How The Arizona Pilot Plan Might Work

The binding constraints for scaling a semiconductor corridor in the U.S. Southwest are reliable water, firm 24/7 electricity, and industrial material inputs - exactly the three outputs CSDP can provide. A CSDP pilot program aligned with advanced semiconductor manufacturing is the fastest path to demonstrating proof-of-concept while attracting useful commercial investment. Semiconductor fabrication facilities and AI data centers are uniquely creditworthy customers for both reliable power and high-quality water.

Phoenix currently receives approximately 36 to 40 percent of its water from the Colorado River, delivered through the Central Arizona Project (CAP) canal. It is pumped 336 miles uphill from Lake Havasu. Phoenix electric power currently comes from a mix of nuclear, natural gas, coal, hydro, and solar.

Arizona's proximity to the Sea of Cortez (~300 miles) supports a practical triad: a coastal CSDP hub producing pipeline conveyance of water to the Arizona semiconductor corridor, high-voltage transmission of electrical power, and critical rare minerals needed for chip manufacturing. This project would result in long-term investment agreements among semiconductor fabrication facilities (TSMC), AI computer chip designers (Nvidia), and Arizona state utilities. This converts first-of-a-kind risk reduction into contracted infrastructure finance.

6.5 Corridor Alignment and Deployment Sequencing

Two physically and institutionally realistic alignments dominate technical and policy planning for a Gulf of California intake and coastal CSDP complex. These alignments serve different roles in a phased strategy rather than representing mutually exclusive alternatives.

1. The Yuma - San Luis Río Colorado corridor integrated with the Colorado River / CAP system.
The Yuma - San Luis Río Colorado alignment integrates with the existing CAP system through a binational exchange structure. Desalinated water would serve coastal demand while Arizona receives augmentation through CAP. This minimizes a new right-of-way in Arizona, creates immediate access to CAP as a credit-grade electricity producer and produces the shortest path to financial self-sufficiency.

2. The Puerto Peñasco - Sonoyta - Lukeville - Central Arizona corridor
The Puerto Peñasco alignment enables a direct export configuration in which the coastal CSDP plant produces 24-hour baseload electricity, thermally distilled water, and mineral commodities for conveyance to the Phoenix Active Management Area. This configuration supports a binational power node, a port-linked industrial cluster, and long-term global replicability. Its first-stage constraints are new cross-border pipelines, higher total lift, and corridor permitting agreements between Mexico and the United States.

6.6 Proposed Revenue Structure and Financial Risk Reduction

CSDP plants would be financed as a baseload power utility with water and minerals as by-products. Electricity under long-term contracts would form the primary credit anchor. Water is monetized as a reliability service through augmentation agreements. Mineral recovery provides inflation-linked commodity exposure and supports a coastal industrial hub. This diversified structure reduces the weighted average cost of capital, stabilizes tariffs, protects against market volatility, and increases equity valuation without increasing debt risk.

6.7 Recommended Sequencing Strategy

Phase A is structured for private capital deployment through the Yuma / CAP integration model. It uses anchor power electricity, avoids long-distance product pipelines, and demonstrates real unit economics under standardized construction.

Phase B expands through the Puerto Peñasco export configuration. The Arizona Pilot Project then becomes a global reference plant for standardizing a system of CSDP installations, enabling global scale deployment at mass production levels that will drive down capital investment costs and increase profitability.

One large CSDP plant would produce more water and electricity than TSMC would need. The excess would be sold to satisfy shortages in the surrounding areas.

7. Potential Sea Level RiseMitigation

7.1 Threat to the Sovereignty and Territorial Integrity of Many Nations.

Sea level rise represents one of the most direct threats to the sovereignty and territorial integrity of many nations, particularly Small Island Developing States (SIDS) and low-lying coastal nations. Current scientific data confirms that the rate of rise has more than doubled in three decades, from 2.1 mm per year in 1993 to 4.5 mm per year by 2023, with the year 2024 recording an exceptional rate of 5.9 mm, driven largely by thermal expansion of warming oceans and glacier melt.

To hold sea levels at their current elevation would require the permanent removal of approximately 1.62 trillion cubic meters of water from ocean circulation annually - a volume equivalent to one-seventh of Lake Superior, every year, indefinitely. IPCC projections for end-of-century sea level rise range from 0.4 to 1.0 meters under moderate scenarios, with extreme ice-sheet collapse scenarios exceeding 2 meters.

The consequences are not abstract:

1. Permanent inundation of Pacific and Indian Ocean island nations, including Tuvalu, Kiribati, and the Maldives.

2. Displacement of an estimated 200 to 400 million coastal inhabitants by 2100 under high-rise scenarios.

3. Salinization of agricultural deltas in Bangladesh, Egypt, Vietnam, and Myanmar, threatening food security for hundreds of millions.

4. Loss of critical coastal infrastructure : ports, cities, and power plants representing trillions of dollars in sovereign assets.

5. No currently endorsed international program provides a concrete mechanism to physically remove water at large scale from the world's oceans.

There is currently no program in existence or even in planning stages being planned that could substantially reduce the looming threat of sea level rise. But at some time in the future, the existential threat may force a re-evaluation of what is possible, or what must be done.

Would a global network of CSDP plants stop sea level rise? No! To offset the current sea level rise estimate of 4.5 mm/year, approximately 4.44 billion cubic meters of water must be removed from ocean circulation daily. A network of approximately 4,400 CSDP plants, each processing an average of 1.03 million m^3/day, would be needed to even come close that target. Even then, rise would slow, not stop. Further rise would be compounded by continued green-house gas emissions.

Is a network of 4,400 plants this scale feasible No! Certainly not at this time. Even if the plants worked as advertised, until sea level rise reaches catastrophic proportions, the international community would never have the will to undertake a such a program.

The 4,400 plant target is used only for planning purposes and the worst case cost scenario. Building 4,400 plants that remove 1.03 million m^3/day should significantly reduce sea level rise. However, the practical and maximum daily water processing capability for economical operation needs to be determined, and it is also recognized to be an enormous cost, and possibly an unachievable engineering feat.

Because of the tremendous flexibility in locating these plants, coupled with the economy of scale benefits of mass production, deploying more smaller plants with less desalinization capability may be a better option. A smaller number of plants, say 2,200, or smaller plants extracting less sea water, could still reduce sea level rise, buying time for nations to develop additional ways to halt the rise.

CSDP plants should be understood as as one pillar of a comprehensive response to an existential threat. They would complement any other plans to counter sea-level rise nations may develop, not replace or compete with them. Global CSDP installations of that magnitude are currently impossible to achieve. However, taking steps to develop the concept into a technological viable, financially profitable system now just might provide an valuable insurance policy needed in the future.

The sun rises every day. The ocean rises every year.
One of these may someday be put to work stopping the other.

7.2 Inland Water Fate: A Hypothesis to Validate Through Field Measurement

If CSDP is used to move desalinated water inland to attempt to slow sea level rise, a question would be won't water transported inland "return to the ocean anyway?" Might evapotranspiration, runoff, and river discharge reduce or negate any sea-level mitigation benefit? Rather than treat inland retention as a settled value, this program treats it as a measurable parameter that must be validated empirically at pilot scale.

The working hypothesis used for planning purposes is that, in arid and semi-arid basins with substantial soil moisture deficit, porous aquifers, and managed groundwater recharge, a large fraction of delivered water will remain stored on land over human-relevant timescales (months to decades). The paper's preliminary planning assumption is that net inland retention would be high, on the order-of-magnitude of 90 percent depending on basin characteristics and water management practices. Analyses based on global hydrological modeling literature - including frameworks developed by Hanasaki et al. (2023) and Gao et al. (2023), indicates that in arid and semi-arid environments, short-term inland water retention might exceed 95 percent over human-relevant timescales. However, the exact retention rate is site-specific and cannot be responsibly asserted as universal without field data. Here are the initial optimistic estimates of inland water retention:

Water Fate	Share of Total	Effect
Soil moisture retention	35–50%	Supports agriculture, prevents erosion

Evapotranspiration	20–40%	Drives microclimate formation, increases rainfall
Vegetation uptake	10–25%	Carbon sequestration, ecological restoration
Aquifer recharge	5–20%	Long-term water security
Surface runoff to ocean	<5%	Negligible return to sea

The evapotranspiration pathway, representing 20 to 40 percent of distributed water, often mischaracterized as 'lost' water is in fact the most climatically valuable effect that acts as a carbon sequestration mechanism. In fact, water vapor driven inland increases local humidity, promotes cloud formation, and ultimately generates precipitation over land rather than ocean, initiating a self-reinforcing greening cycle. Increased vegetation in turn captures and stores atmospheric carbon dioxide, contributing directly to climate mitigation goals.

The Sahara greening hypothesis, supported by multiple peer-reviewed studies, confirms that large-scale irrigation of arid land measurably reduces regional temperatures and increases rainfall over multi-decadal timescales. This hydrological retention model is a theoretical balance model derived from global hydrological literature and could be validated through pilot plant field data.

Accordingly, the CSDP pilot program will produce the first standardized, open, basin-scale dataset capable of quantifying: (a) how much delivered water becomes long-lived groundwater storage, (b) how much becomes soil moisture and vegetation storage, (c) how much returns to the ocean via runoff, and (d) how much cycles through the atmosphere and where that precipitation falls.

7.3. The Inland Water Fate Measurement and Verification (M&V) Program

To evaluate the inland water fate hypothesis, each pilot deployment will be asked to include a dedicated Measurement & Verification (M&V) program designed to estimate net retention with defensible uncertainty bounds.The M&V program should estimate the partition of delivered water into the following pathways:

- Aquifer recharge / managed groundwater storage (long-duration retention).
- Soil moisture storage and agricultural uptake.
- Vegetation growth and biomass storage (including carbon co-benefits).
- Evapotranspiration (ET) flux to the atmosphere.
- Surface runoff and river discharge to the ocean.
- Operational losses (leaks, evaporation from canals/reservoirs, treatment losses).

The measurement stack should combine standard hydrology methods used by water agencies and climate science:

- Metered delivery volumes at pipeline and distribution nodes (mass balance baseline).
- Groundwater monitoring wells (water table change and storage coefficients).
- Reservoir/canal level monitoring and evaporation estimation.
- Remote sensing estimates of ET and vegetation response.
- (cross-validated with ground stations).
- Stream gauges / discharge estimates on relevant watershed outlets.
- Isotopic or tracer studies.
- (optional) to distinguish new delivered water from background flows.
- A basin-scale water balance model calibrated to observed data.

Each pilot would publish an annual retention report including:

- Estimated net inland retention fraction with confidence intervals.
- Sensitivity of retention to basin type, seasonality, irrigation method, and recharge strategy.
- Time-to-return distribution (what fraction returns within 1 year, 5 years, 20 years).
- Implications for sea-level mitigation.

This converts the retention question from an argument into a bankable engineering uncertainty that can be narrowed over time.

8. Summary of Anticipated Concerns

A number of initial concerns have to be overcome to assure government entities, investors, engineering firms, and skeptics of the viability of the project.

8.1 Technology readiness?

Each CSDP component has been demonstrated to be operationally possible separately, either commercially or by pilot projects. No breakthrough inventions are required. The innovation required is integration of existing technologies and how they might be brought up to global scale through mass production. However, the next step is for engineering firms familiar with these technologies to thoroughly review the concept for practical application.

To clearly understand the feasibility, time lines, and economics of this idea, interested government entities and investment organizations could issue a Request For Proposal (RFP) from engineering firms having experience in design, production, and construction of large-scale power, solar, and desalinization plants. The RFP should demonstrate how the firms would integrate the CSDP technologies into a unified system.

8.2 Capital cost?

The initial plants will be a high in absolute costs, but manageable when amortized over 20 years. Mass production is expected to reduce some costs 60 to 75 percent, and revenue streams produced will help pay for the program. However, economic analyses by an objective, reputable financial firm are needed to solidify projected costs and profits to assure investors.

8.3 Water distribution infrastructure?

Inland pipeline networks add cost and complexity, but water revenue funds can help finance pipeline build-out. Realistic Economic and ecological benefits projected for host nations must be examined and documented honestly.

8.6 Evaporation returns water to ocean anyway?

Hydrological models presented in this paper indicated only a fraction of evapotranspiration eventually reaches ocean via the atmosphere. They have suggested as much as 95 percent of CSDP water would be retained on or in the land. Further research and serious computer modeling as outlined above is needed to validate those assumptions.

9. Conclusion: Why CSDP? Why Now?

Concentrated Solar Desalination Power plants are not a pie-in-the-sky hypothesis. Much of the technology operates commercially today. What has been missing is the realization that deploying the unified CSDP system can create a self-financing infrastructure network with a humanitarian and environmental impact unmatched by any prior human project. The addressable markets for fresh water, cleanly produced continuous electricity critical minerals are immense. The shortages are going to continue into the foreseeable future and represent an opportunity worth trillions of dollars to visionary corporations. The first-mover has a real advantage. Immediate standardization lowers costs and creates a

protective moat for investors. Creating a global installation capability is one of the largest business opportunities of the 21st century.

In addition to sea-level mitigation, CSDP supplies the immediate physical inputs required for strategic safeguards of production of AI data centers and advanced semiconductor manufacturing, turning resilience into revenue, redundancy into bankable infrastructure and geopolitical fear into a strategic safeguard. The Arizona pilot project makes the TSMC backup plant goals much more achievable and quicker, reducing global tensions and digital insecurity.

The global community has spent three decades debating climate change and produced no scalable, concrete engineering solution to sea level rise. Time is running out. The options on the table - carbon taxes, lifestyle adjustment, solar panels, wind farms, and international pledges do not move useable water from the ocean to the land. A competing unified technological approach is non-existent. No current competitor addresses all three pressing problems of water, electrical, and mineral shortages in one unified environmentally friendly approach. This proposal does. If CSDP plant are efficiently engineered, when the coming sea level catastrophe arrives, the world, CSDP will be available to help mitigate the crisis.

The question remains: if the technology is proven, the power source is free, the economics are favorable, and the need is existential - why isn't a pilot project underway?

The answer lies not in science or technology, in finance, or resources, but in our imagination, our political will for survival and our commitment to upholding humanitarian ideals.

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This document integrates Perelli's original proposal with revised quantitative analysis, current sea level science, and investment framing. The hydrological retention model referenced herein is a theoretical balance model derived from global hydrological literature and should be validated through pilot plant field data.